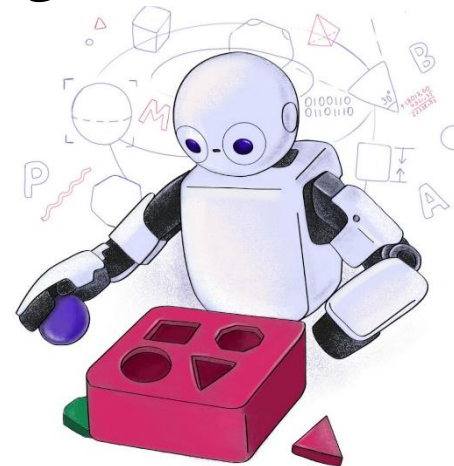


TP557 - Tópicos avançados em IoT e Machine Learning: *Motion Classification*



Inatel

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samuelbmafra@inatel.br

Motion Classification



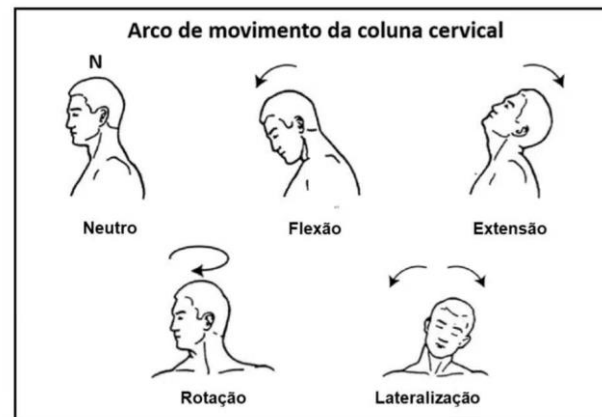
Queda de dispositivos



Controle logístico



Monitoramento de animais



Exercícios físicos, movimento de cadeira de rodas

Motion Classification

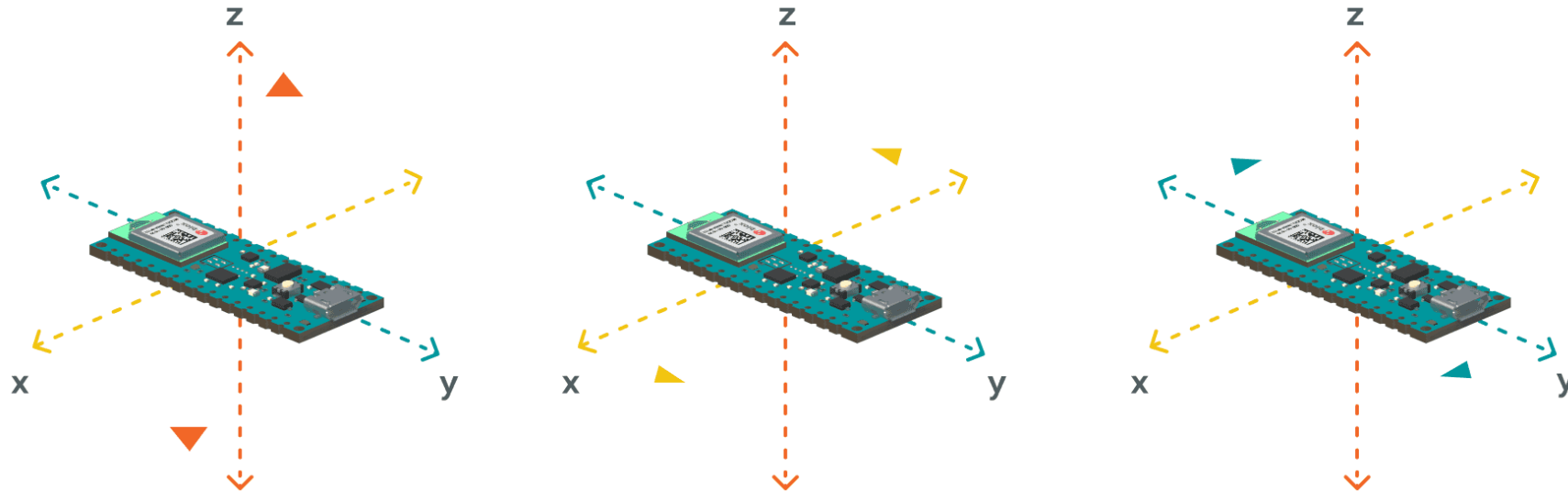
- Movimentação de containers ao longo do ciclo logístico usando um acelerômetro.



Motion Classification

- Movimentação de containers ao longo do ciclo logístico usando um acelerômetro.

Um acelerômetro é um dispositivo usado para medir forças de aceleração. O **LSM9DS1** mede a aceleração em 3 eixos.



Motion Classification

- Container parado no porto:



Movimentação em nenhum eixo.

Motion Classification

- Container sendo erguido:



Movimentação no eixo z.

Motion Classification

- Container transportado no oceano:



- Movimentação em várias direções (x,y,z) devido às correntes marítimas.

Motion Classification

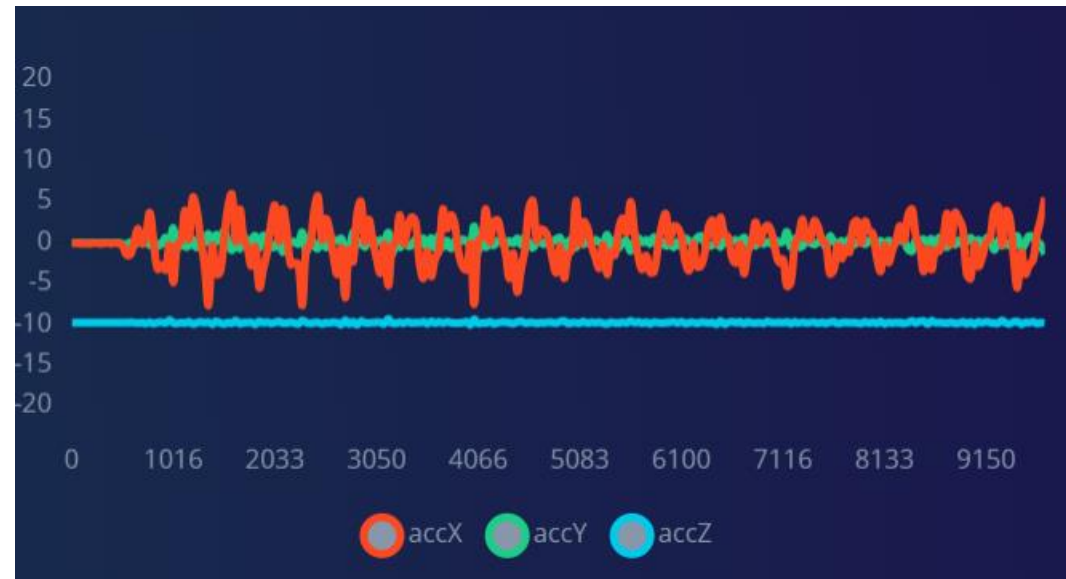
- Container transportado no caminhão:



- Movimentação no eixo x.

Motion Classification

- Exemplo de amostra coletada com acelerômetro:



- Movimentação no eixo x.

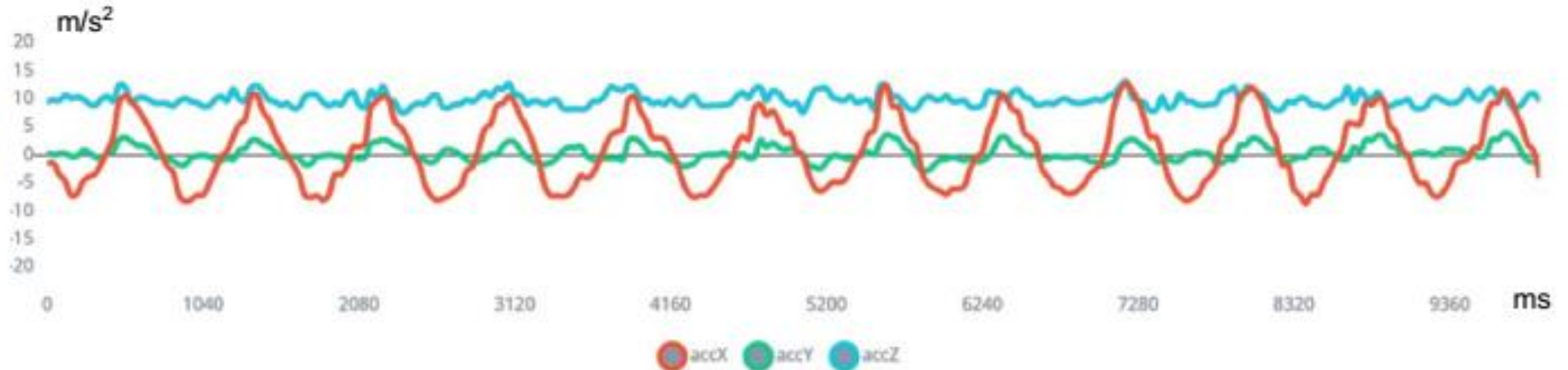
Motion Classification

Aplicação: Classificação de 4 estados do container

- Parado
- Sendo erguido
- No mar
- No caminho

Motion Classification

- Exemplo de 10 segundos de dados coletados por um acelerômetro com frequência de leitura de 62.5 Hz, i.e. 62.5 amostras por segundo.
- Que tipo de dado é este?



Motion Classification

- **Série temporal**: cada amostra tem *correlação* com as anteriores e posteriores.
- Assim, por serem séries temporais, vamos *processar janelas que capturem pelo menos um ciclo de movimento*, por exemplo, 2 segundos.



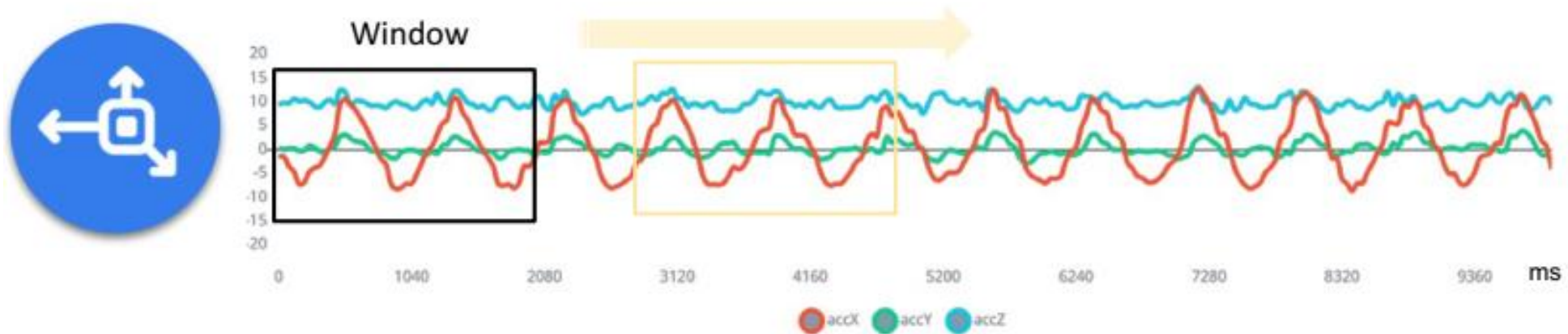
Motion Classification

- Que tipo de rede devemos usar?

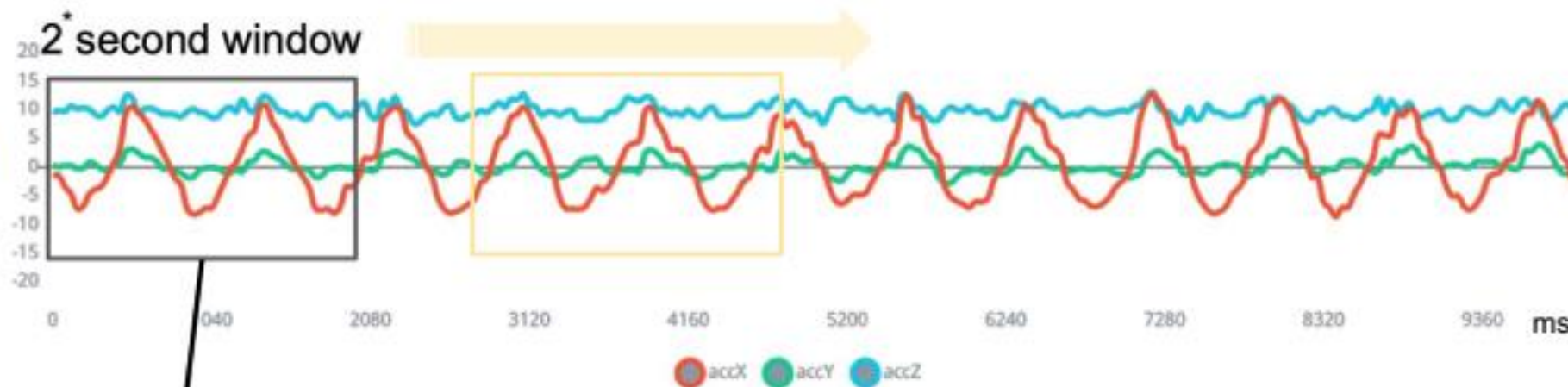


Motion Classification

- **Redes com memória, para que elas se lembrem do passado!**
- Redes baseadas em camadas do tipo Long Short-Term Memory (LSTM) e Gated Recurrent Units (GRU), por exemplo.



Motion Classification



Raw Features as a window

- 125** samples for each axis (62.5Hz x 2s)
- 375 total features (125 x 3 axis)

375 atributos de entrada!

* 2 seconds is needed to capture 1 or 2 cycles of movement

** 2 seconds at sample rate of 62.5 Hz -> 125 samples

Motion Classification



Raw Features as a window

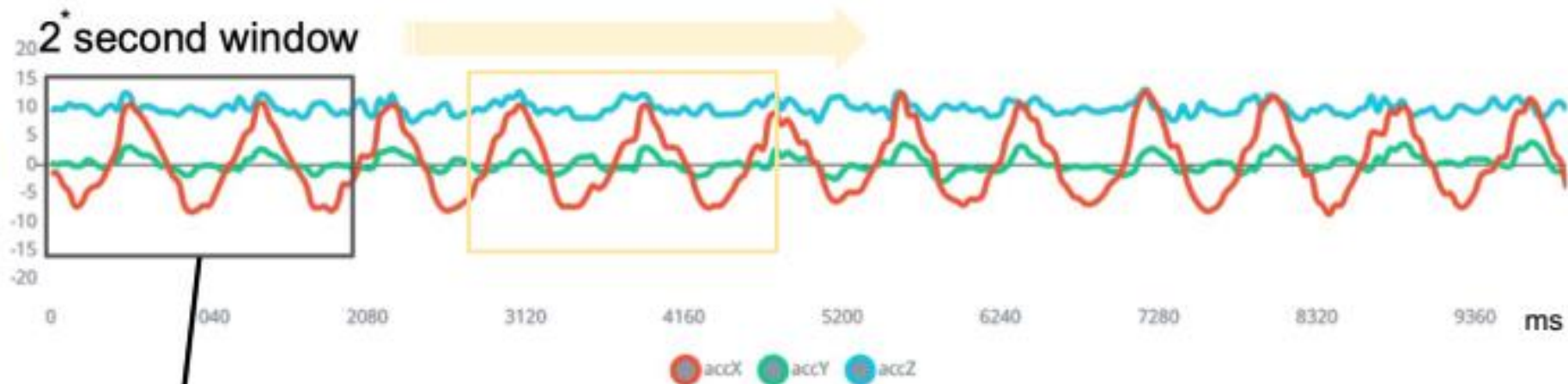
- 125** samples for each axis (62.5Hz x 2s)
- 375 total features (125 x 3 axis)

* 2 seconds is needed to capture 1 or 2 cycles of movement

** 2 seconds at sample rate of 62.5 Hz -> 125 samples

*Precisamos de uma rede complexa (e.g., LSTM) para processar todos os atributos de entrada e **compreender os dados de cada janela.***

Motion Classification



Raw Features as a window

- 125** samples for each axis (62.5Hz x 2s)
- 375 total features (125 x 3 axis)

* 2 seconds is needed to capture 1 or 2 cycles of movement

** 2 seconds at sample rate of 62.5 Hz -> 125 samples

A consequência é que o modelo consome mais memória e pode não operar em tempo real.

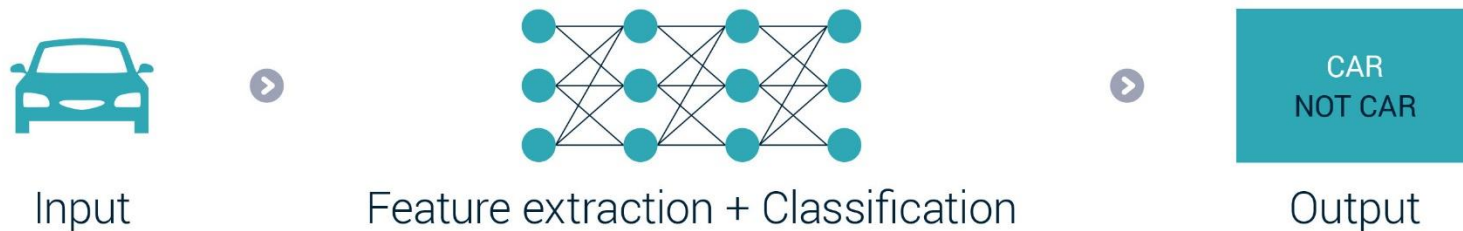
O que podemos fazer?

Motion Classification

Machine Learning

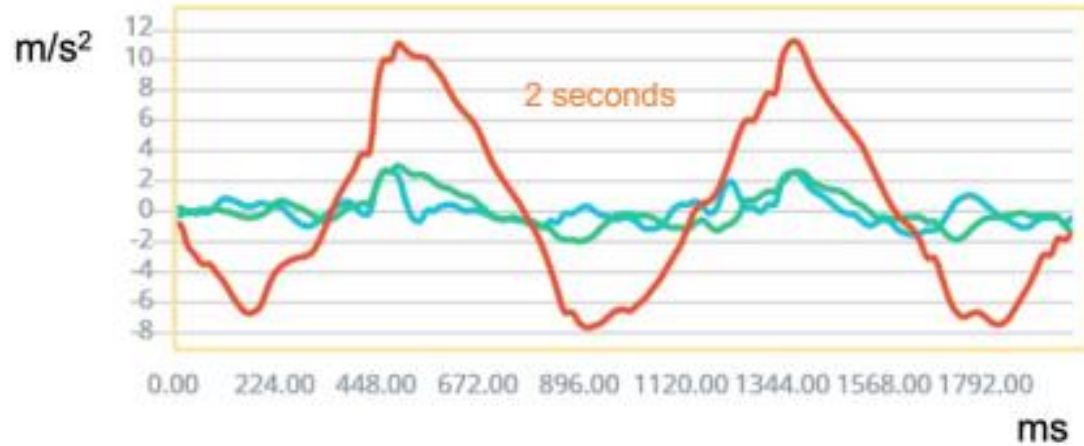


Deep Learning



*Ao invés de termos uma rede muito complexa que faça a **extração de atributos de forma automática**, vamos pré-processar os dados e extrair alguns atributos manualmente.*

Motion Classification



Manual Feature Extraction



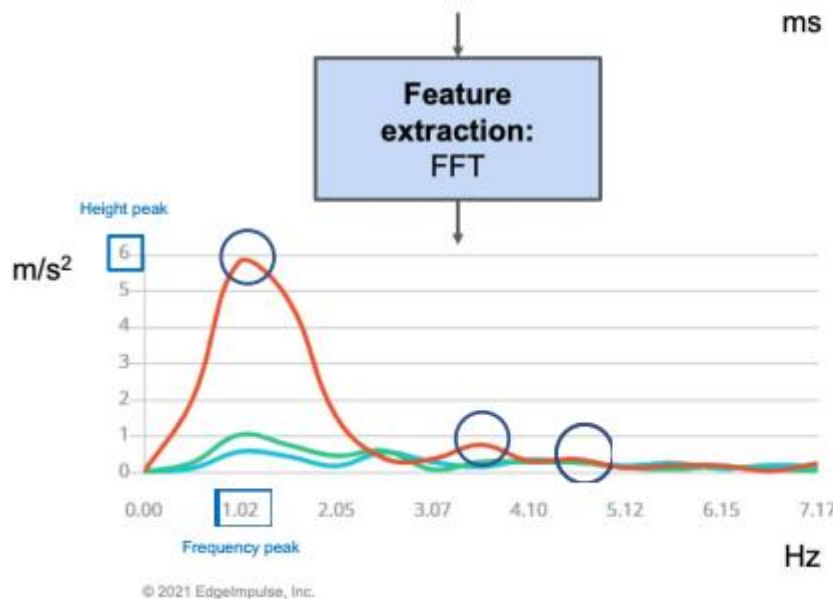
3 RMS (Root Mean Square) values
- one for each axis (x, y, z)

$$x_{\text{RMS}} = \sqrt{\frac{1}{n} (x_1^2 + x_2^2 + \dots + x_n^2)}.$$

125

1. Etapa do pré-processamento:
Calcular o valor RMS de cada um dos eixos.

Inatel CAMINHOS
QUE CONECTAM
COM O FUTURO

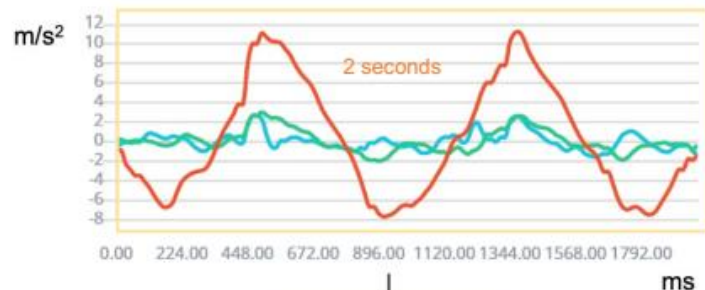


➔ 3 RMS

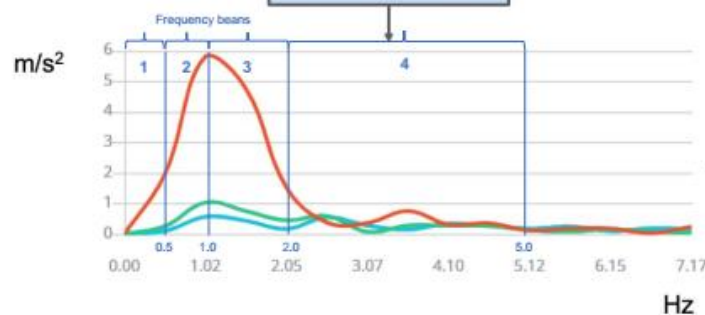
➡ 9 Height + 9 Freq. peak values

Extrair as frequências dos sinais

Motion Classification



Feature
extraction:
FFT



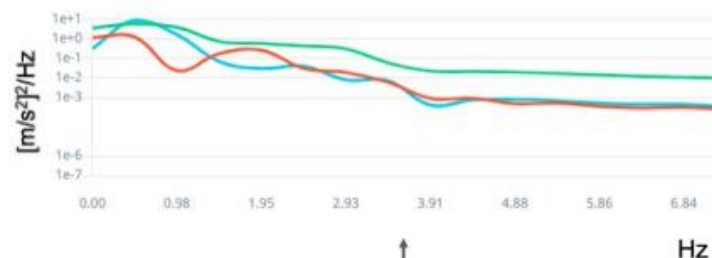
© 2021 EdgetImpulse, Inc.

Manual Feature Extraction



3 RMS + 9 HP + 9 FP + 12 PSD values

Power Spectral Density (PSD)



4 Frequency bins per axis

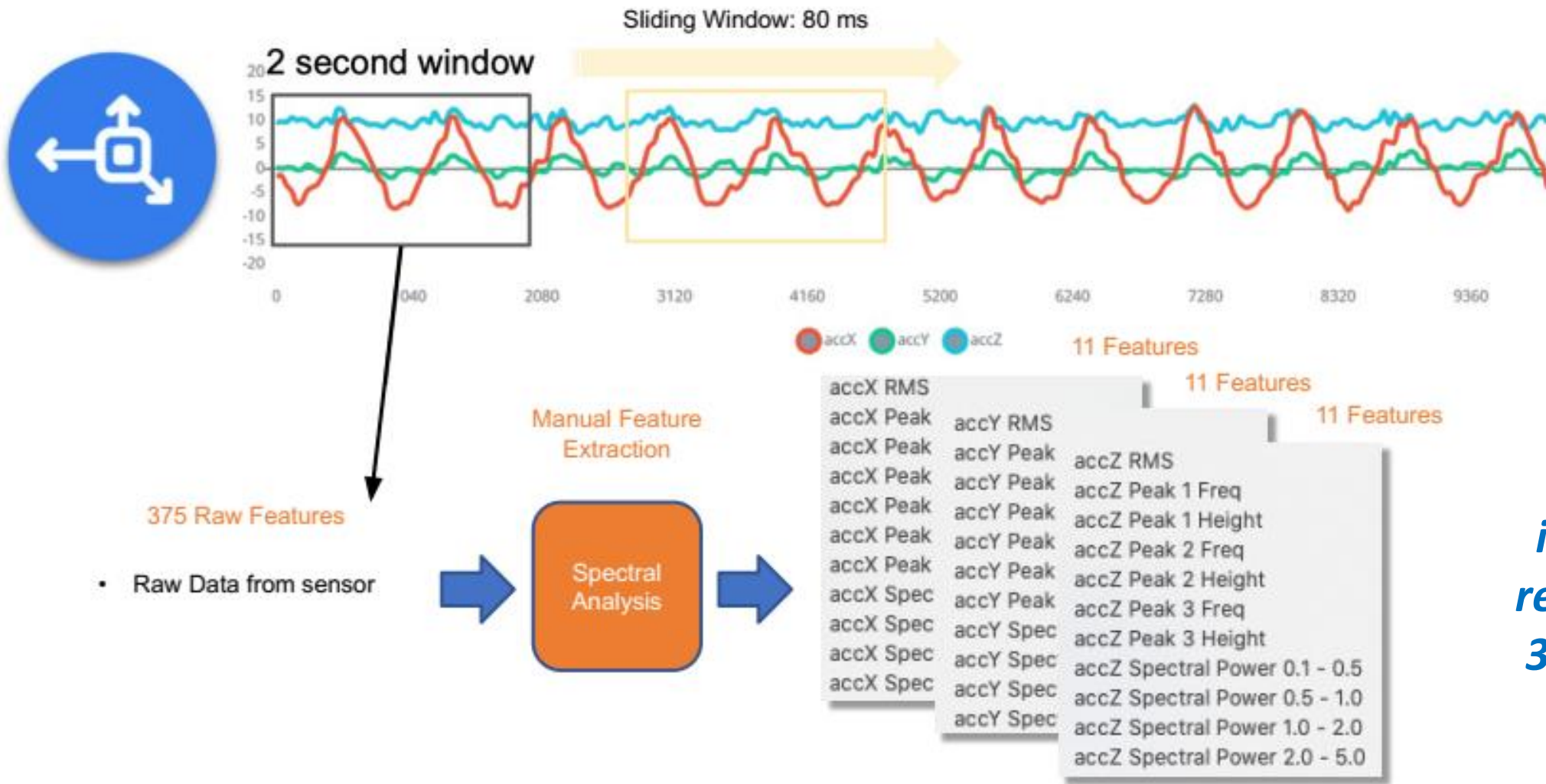
Feature
extraction:
PSD

<https://blog.endaq.com/why-the-power-spectral-density-psd-is-the-gold-standard-of-vibration-analysis>

3. Etapa do pré-processamento:
Densidade espectral de
potência.

Método de Welch: Divide-se a largura de banda em 4 e calcula-se a densidade de potência para cada região. Assim, temos 4 regiões * 3 eixos = 12 atributos.

Motion Classification



Portanto, ao invés de 375 a rede neural terá 33 atributos de entrada.

Motion Classification



A janela de 2 [s] desliza de 80 [ms].



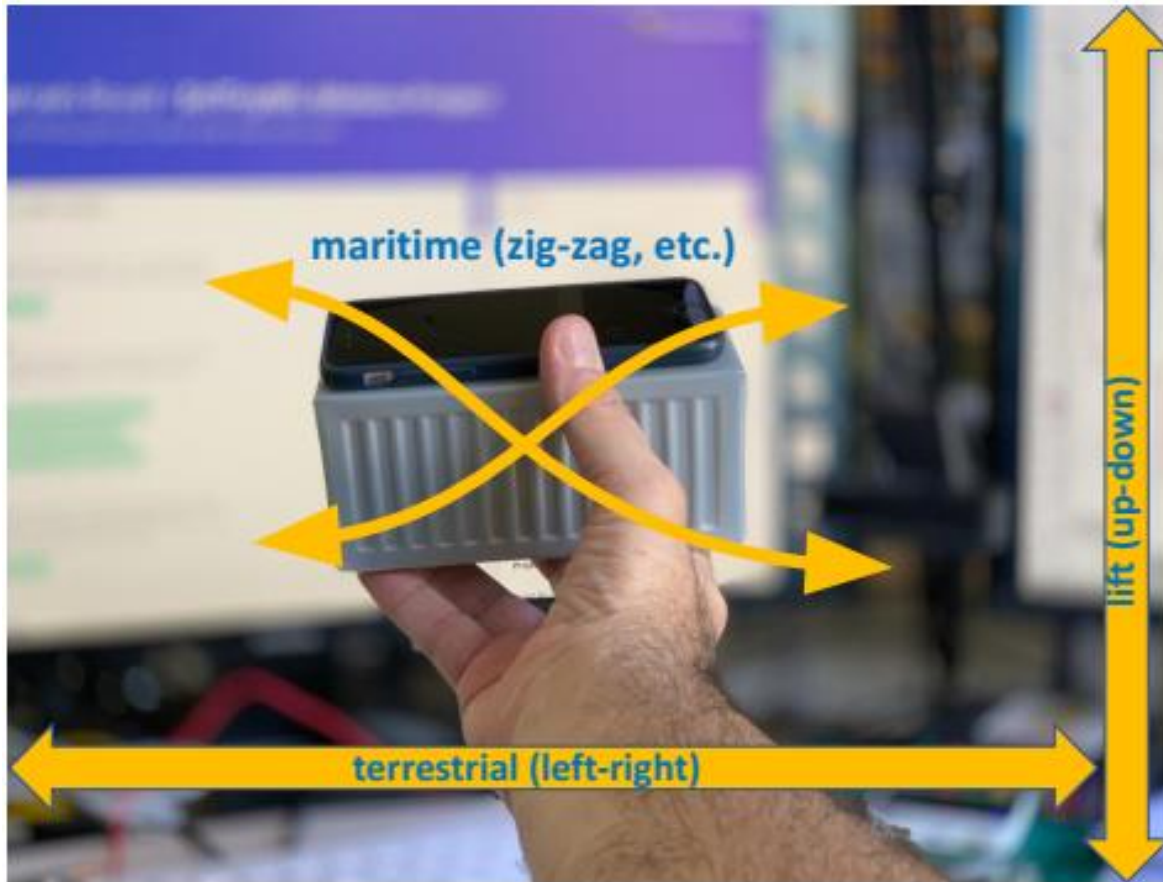
Motion Classification



*Vamos usar o Edge Impulse para criar essa aplicação.
Com ele, vamos desde a coleta de dados, passando pelo
pré-processamento, até a inferência no dispositivo.*

OBS.: O Edge Impulse faz toda a análise espectral (i.e., o pré-processamento) de forma automática.

Motion Classification

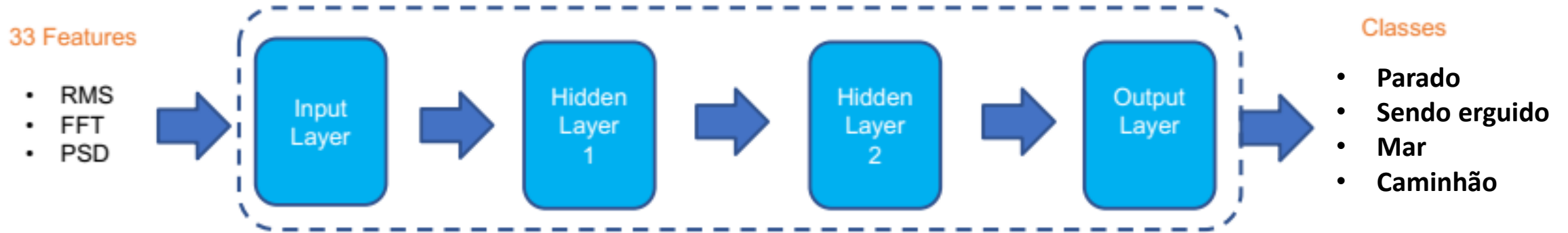


Transportation Classes

- **lift** (up-down)
- **terrestrial** (left-right)
- **maritime** (zig-zag, etc.)
- **idle**

*Vamos criar a base de dados
no Edge Impulse com o
acelerômetro do celular.*

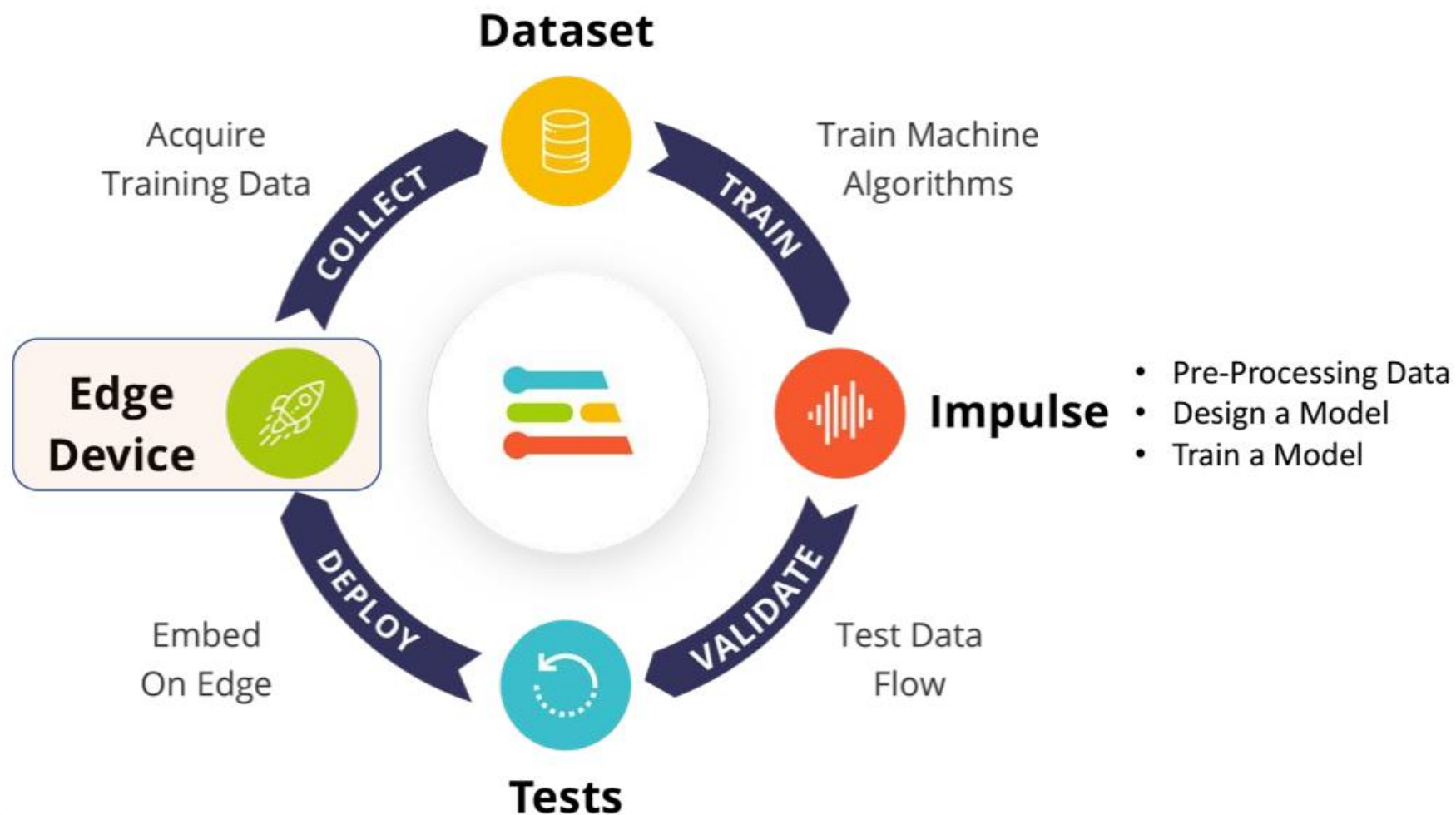
Motion Classification



*Na sequência projetaremos,
treinaremos e testaremos o
modelo.*

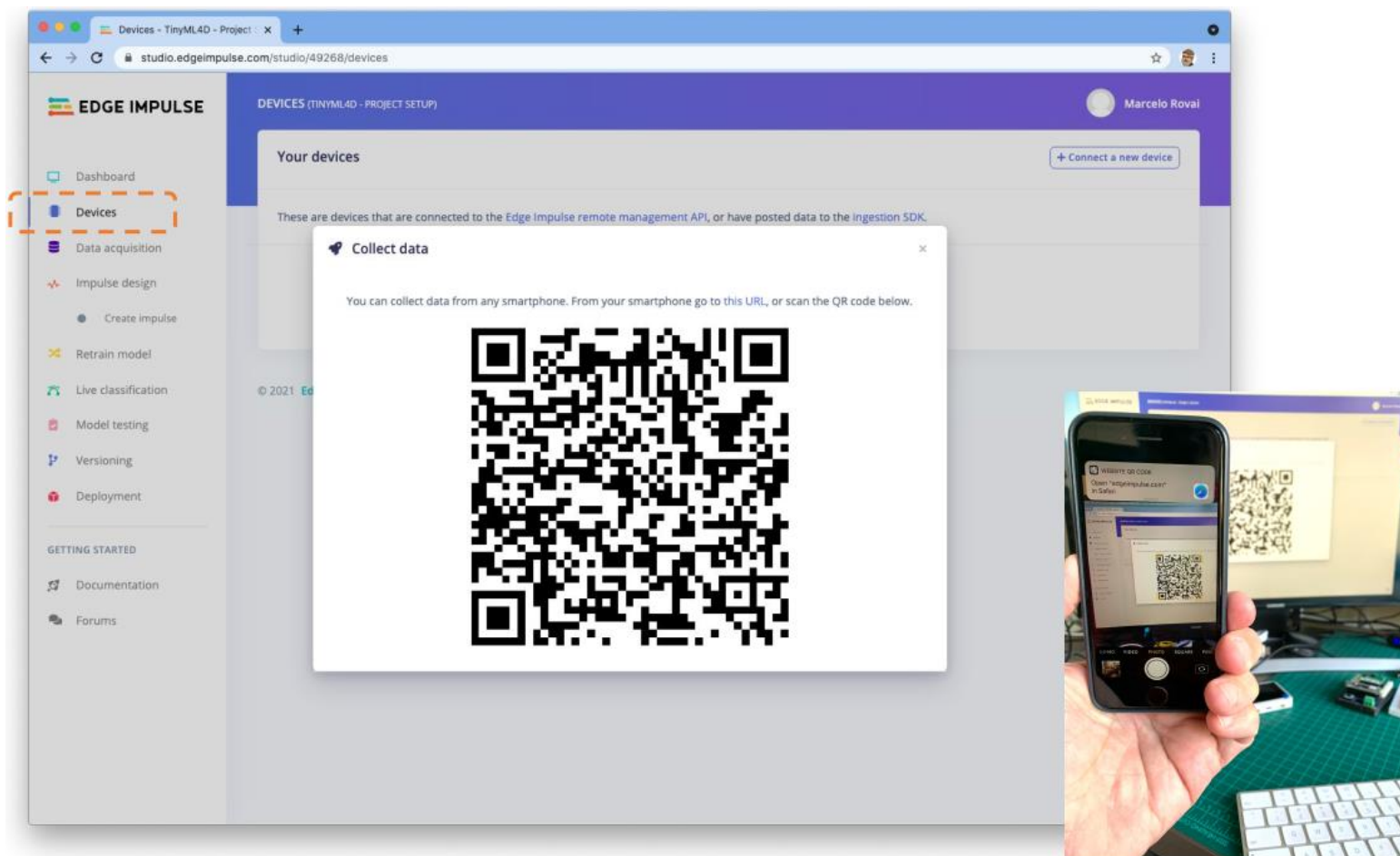
Motion Classification

Passos para criação do classificador via Edge Impulse



Motion Classification

1. Conectando o dispositivo para coleta dos dados



Motion Classification

2. Coleta dos dados

Collect Data

The screenshot displays the Edge Impulse Studio web interface. On the left, a sidebar contains navigation links: Dashboard, Devices, Data acquisition (highlighted with a blue arrow), Impulse design, Create impulse, Spectral Analysis, Neural Network (Ko...), EON Tuner, Retrain model, Live classification, Model testing, Versioning, Deployment, GETTING STARTED, Documentation, and Forums. The main area is titled 'DATA ACQUISITION (SciTinyML-MOTION-PROJECT)' and includes tabs for 'Training data' and 'Test data'. It shows 'DATA COLLECTED 5m 13s' and a 'TRAIN / TEST SPLIT 80% / 20%'. Below this is a table of 'Collected data' with columns for Sample Name, Label, Added, Length, and a status icon. The table lists 15 samples, with the last 5 labeled 'maritime'. On the right, the 'Record new data' panel shows settings for Device (phone_kqfrazik), Label (maritime), Sample length (10000 ms), Sensor (Accelerometer), and Frequency (62.5Hz). A 'Start sampling' button is highlighted with a yellow arrow. Below the settings is a 'Sensor' dropdown menu showing 'Accelerometer', 'Microphone', and 'Camera'. In the bottom right corner, a smartphone screen displays the 'Data collection' app with a circular progress indicator and a '4s' timer.

SAMPLE NAME	LABEL	ADDED	LENGTH
idle.2hstvpk2	idle	Oct 14 2021, 17:54:22	10s
idle.2hstuauf	idle	Oct 14 2021, 17:53:34	10s
idle.2hsttoq3	idle	Oct 14 2021, 17:53:16	10s
idle.2hstt9dk	idle	Oct 14 2021, 17:53:00	10s
idle.2hstsp4a	idle	Oct 14 2021, 17:52:43	10s
idle.2hstrkad	idle	Oct 14 2021, 17:52:06	10s
idle.2hstr3kf	idle	Oct 14 2021, 17:51:49	10s
idle.2hstqaj	idle	Oct 14 2021, 17:51:32	10s
maritime.2hstpkus3	maritime	Oct 14 2021, 17:51:01	10s
maritime.2hsto9ki	maritime	Oct 14 2021, 17:50:16	10s
maritime.2hstnnqu	maritime	Oct 14 2021, 17:49:58	10s
maritime.2hatn60c	maritime	Oct 14 2021, 17:49:40	10s

Motion Classification

3. Criar um impulse

The screenshot shows a web interface for creating an impulse. It is divided into four main panels:

- Time series data (red panel):** Contains settings for 'Axes' (accX, accY, accZ), 'Window size' (2000 ms), 'Window increase' (80 ms), 'Frequency (Hz)' (62.5), and 'Zero-pad data' (checked).
- Spectral Analysis (white panel):** Has a 'Name' field set to 'Spectral Analysis' and 'Input axes' with checkboxes for accX, accY, and accZ, all of which are checked.
- Neural Network (Keras) (purple panel):** Has a 'Name' field set to 'Neural Network (Keras)', 'Input features' with a checked checkbox for 'Spectral Analysis', and 'Output features' set to '4 (idle, lift, maritime, terrestrial)'. There is a 'Save Impulse' button at the bottom right.
- Output features (teal panel):** Shows the output features '4 (idle, lift, maritime, terrestrial)' and a checkmark icon.

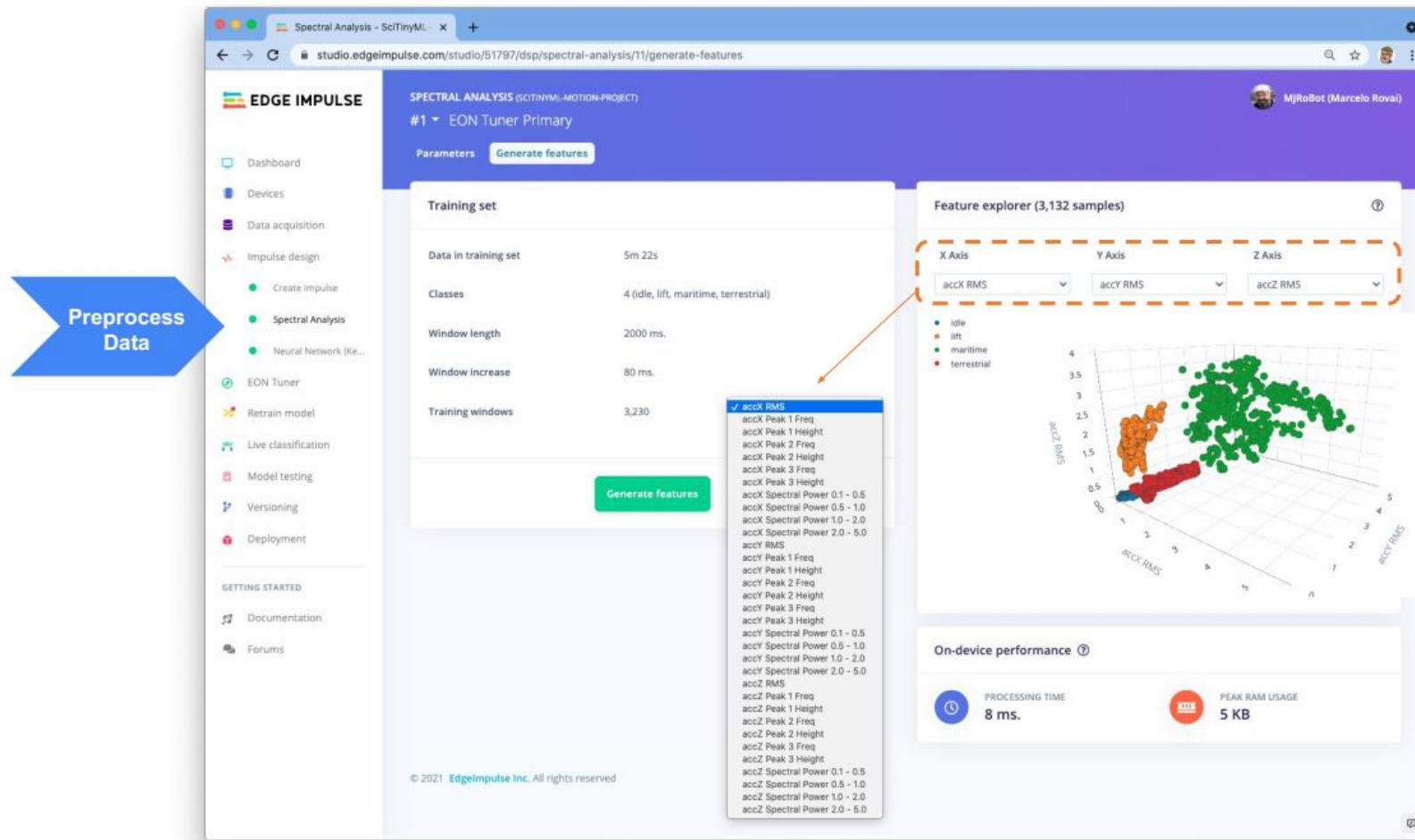


Classes

- Parado
- Sendo erguido
- Mar
- Caminhão

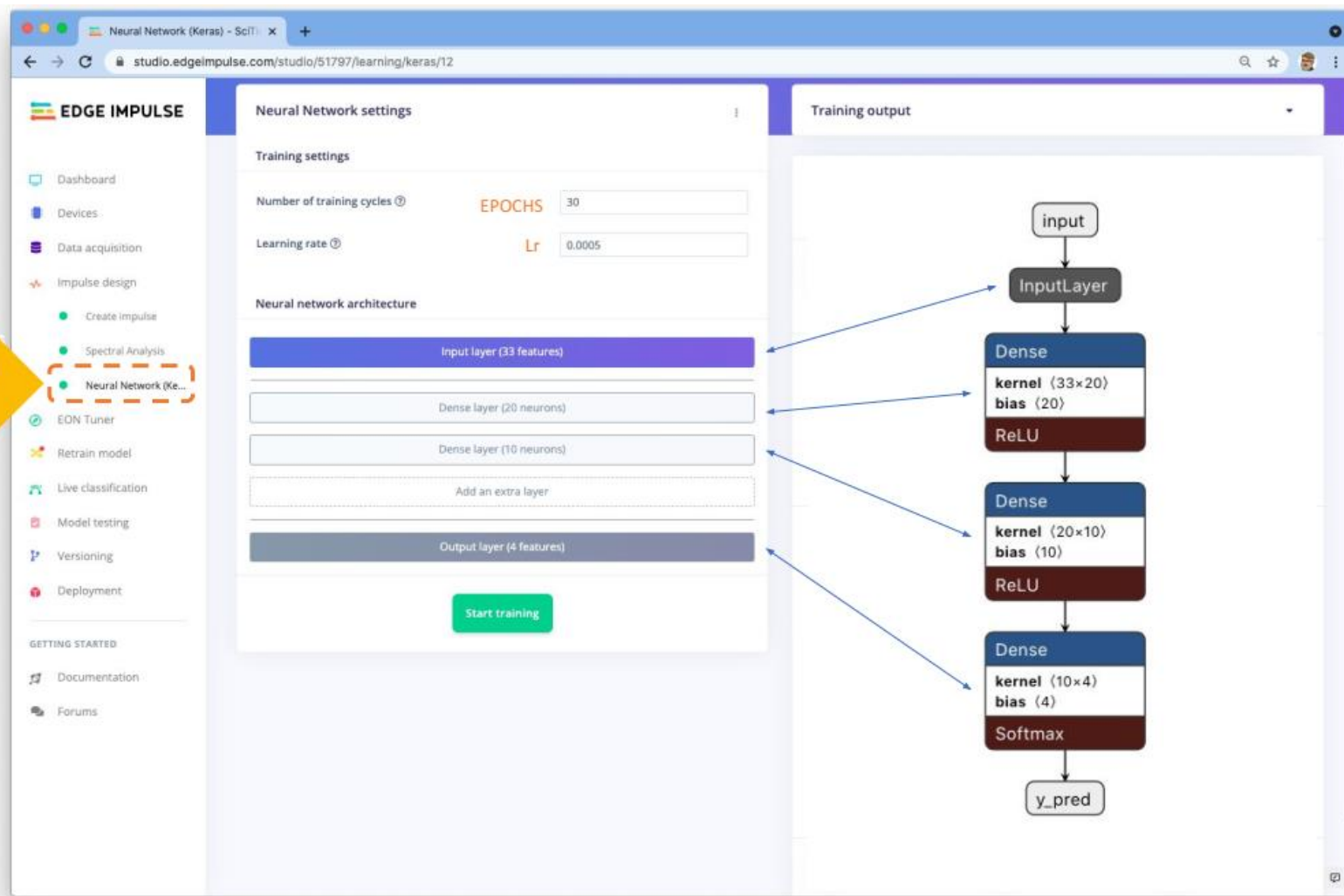
Motion Classification

4. Pré-processamento dos dados



Motion Classification

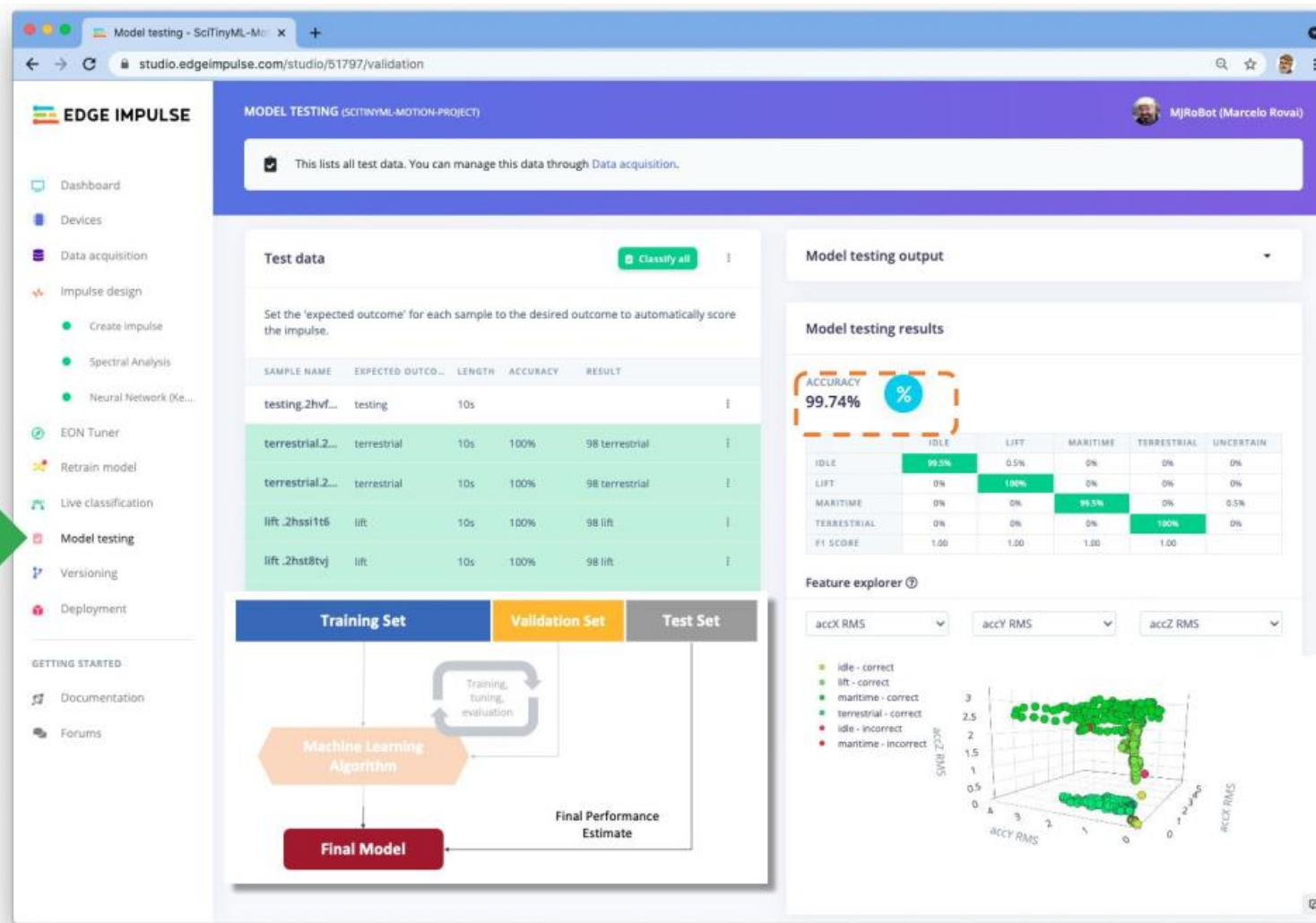
5. Projetar um modelo, treiná-lo e avaliar o resultado



Design a
Model

Motion Classification

6. Avaliar a capacidade de generalização do modelo

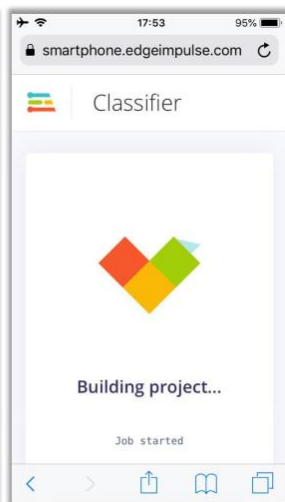
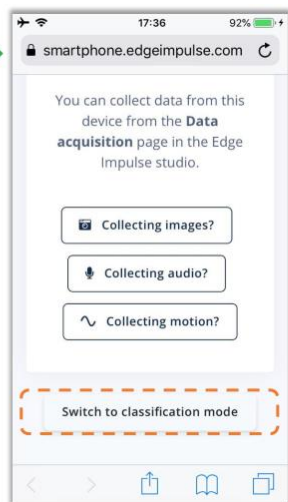


Evaluate
Optimize

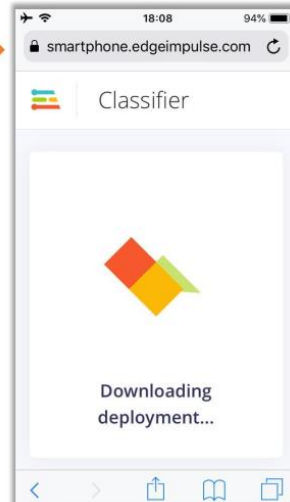
Motion Classification

7. Converter o modelo, fazer o deploy e realizar inferências

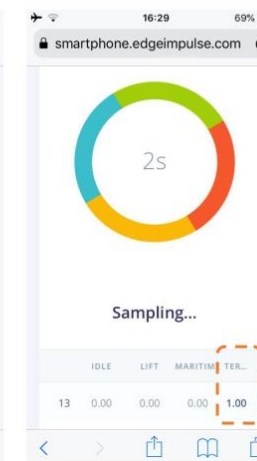
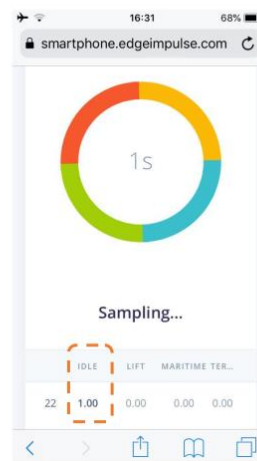
Convert Model



Deploy Model



Make Inferences



Motion Classification

8. Executando no Arduino nano-33 BLE sense

The screenshot displays the Edge Impulse Studio deployment page. On the left, a sidebar contains navigation links: Dashboard, Devices, Data acquisition, Impulse design (with sub-links for Create impulse, Spectral features, and NN Classifier), EON Tuner, Retrain model, Live classification, Model testing, Versioning, and Deployment. Below these are links for GETTING STARTED, Documentation, and Forums. A red arrow points to the 'Deployment' link. The main content area is titled 'DEPLOYMENT (TEST01 - NANO MOTION CLASSIFICATION)'. It features a 'Deploy your impulse' section with a description and a 'Create library' section with a description. Below these are six library options: C++ library, Arduino library (highlighted with a red box), CubeMX CMSIS-PACK, WebAssembly, TensorFlow library, and TensorFlow.js library. The 'Build firmware' section lists various development boards, including ST IoT Discovery Kit, Arduino Nano 33 BLE Sense (highlighted with a red box), Eta Compute ECM3532 AI Sensor, Silabs Thunderboard Sense 2, Himax WE-I Plus, Nordic nRF52840 DK + IKS02A1, Nordic nRF5340 DK + IKS02A1, Nordic nRF9160 DK + IKS02A1, and Nordic Thingy 91. On the right, a 'Build output' panel shows the progress of the build process, including steps like 'Creating job...', 'Writing templates...', 'Copying Edge Impulse SDK...', 'Compiling EON model...', 'Removing clutter and updating headers...', 'Creating archive...', and 'Job completed'.

Motion Classification

8. Executando no Arduino nano-33 BLE sense

The screenshot displays the Edge Impulse IDE interface. On the left is a sidebar with navigation options: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Impulse design. The main workspace shows 'MODEL OPTIMIZATIONS' with a section for 'EON™ Compiler' and options for 'Quantized (int8)' (Selected) and 'Unoptimized (float32)' (Select). A modal dialog box titled 'Built Arduino library' is centered on the screen, featuring a green checkmark icon. The dialog contains the following text: 'Add this library through the Arduino IDE via: Sketch > Include Library > Add .ZIP Library...'. Below this, it states 'Examples can then be found under: File > Examples > anomaly_inferencing'. The background interface is dimmed, showing a 'Build' button at the bottom and a log of operations on the right side.

EDGE IMPULSE

Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design

Create Impulse
Spectral features
Classifier
Anomaly detection
Retrain model
Live classification
Model testing

Upgrade Plan

MODEL OPTIMIZATIONS
Model optimizations can increase on-device performance but may reduce accuracy.

EON™ Compiler
Same accuracy, 54% less RAM, 60% less ROM.

Quantized (int8)
Selected ✓

Unoptimized (float32)
Select

To compare model accuracy, run

Estimate for Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz) [Change target](#)

Built Arduino library

Add this library through the Arduino IDE via:
Sketch > Include Library > Add .ZIP Library...

Examples can then be found under:
File > Examples > anomaly_inferencing

Build

Job scheduled at 22 May 2025 14:35:39
Job started at 22 May 2025 14:35:42
Writing templates...
Writing templates OK

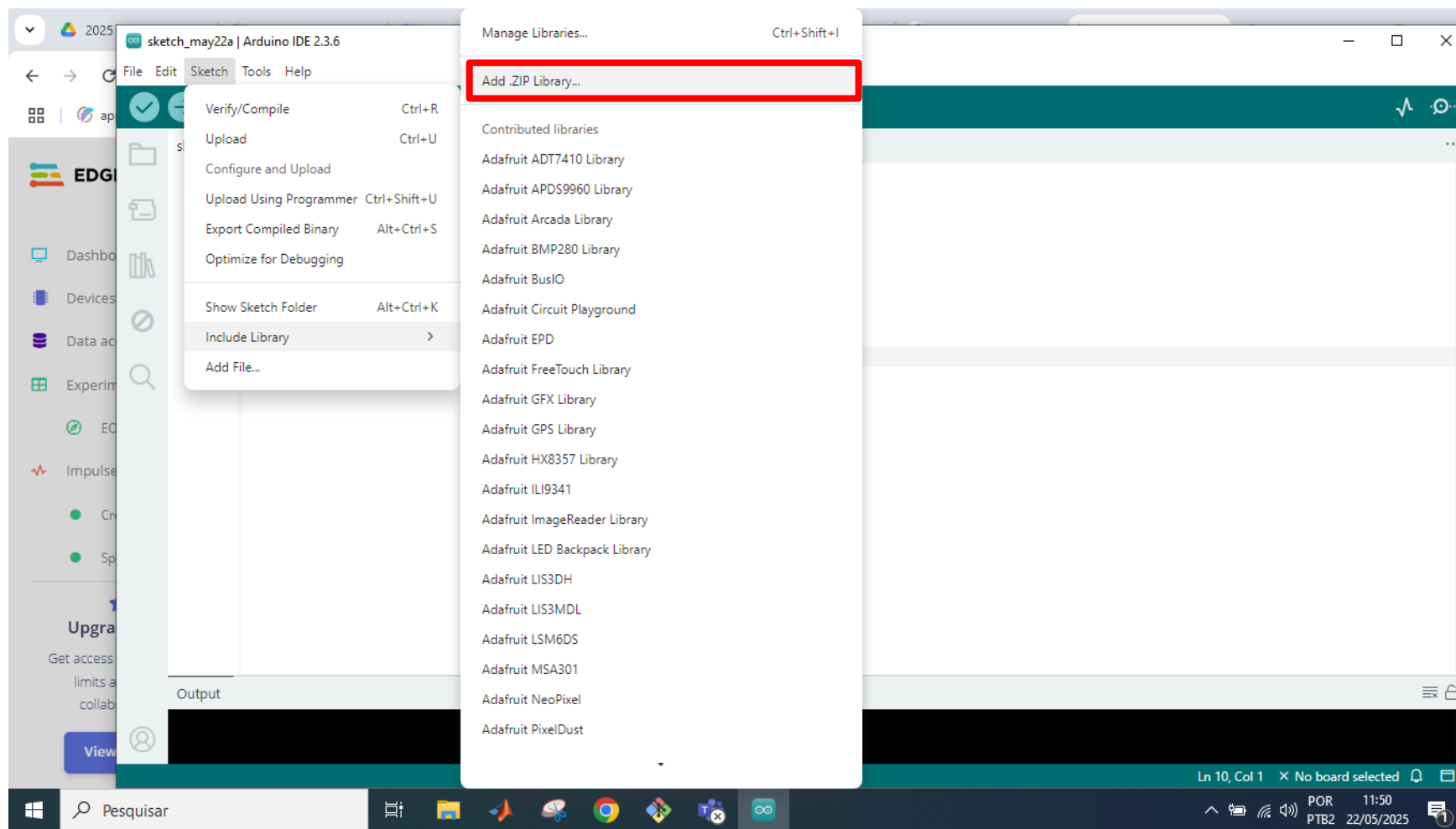
Job scheduled at 22 May 2025 14:35:52
Job started at 22 May 2025 14:35:53
Copying Edge Impulse SDK...
Copying Edge Impulse SDK OK

Compiling EON model...
Compiling EON model OK

Removing clutter and updating headers...
Removing clutter and updating headers OK

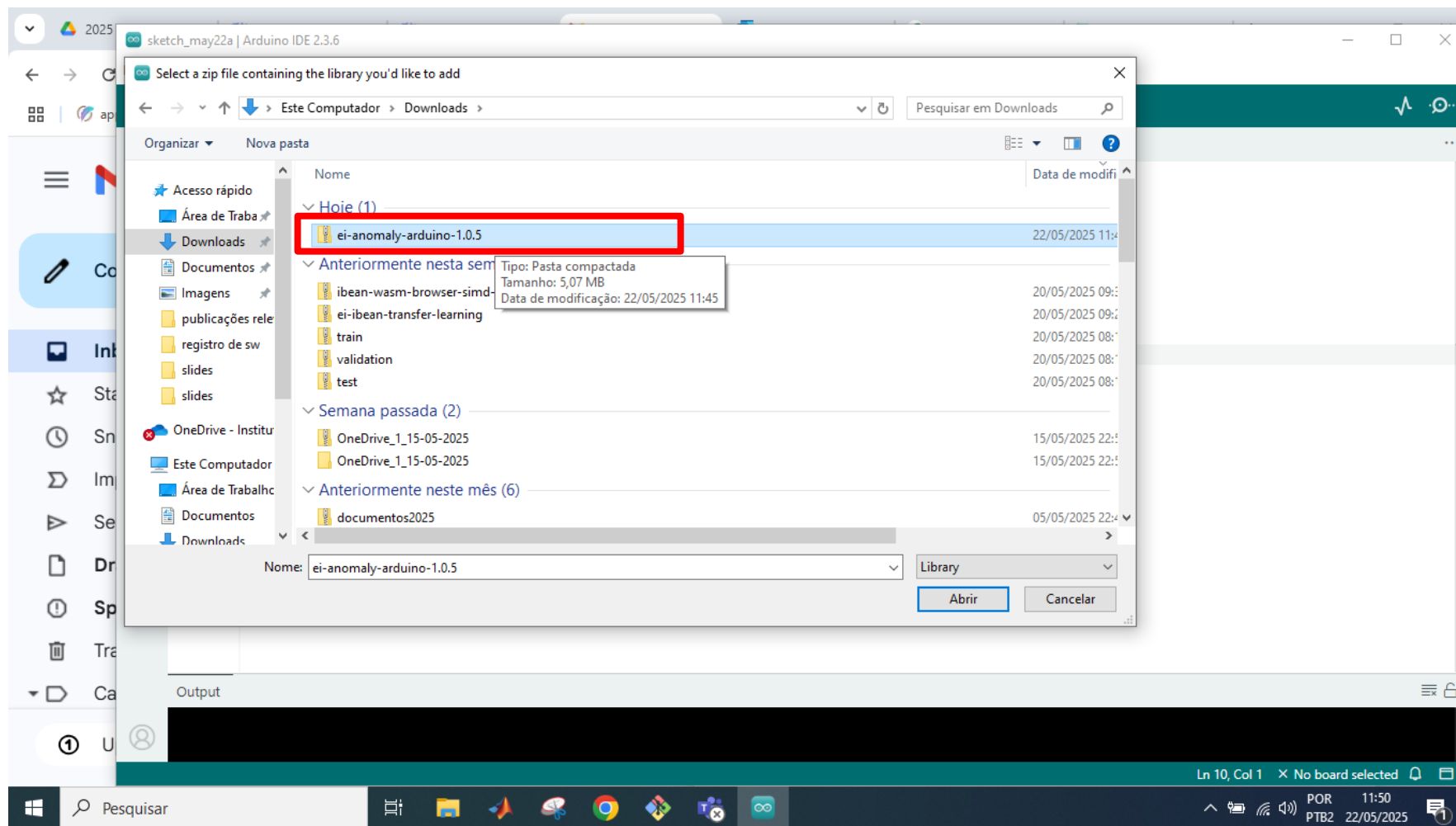
Motion Classification

8. Executando no Arduino nano-33 BLE sense



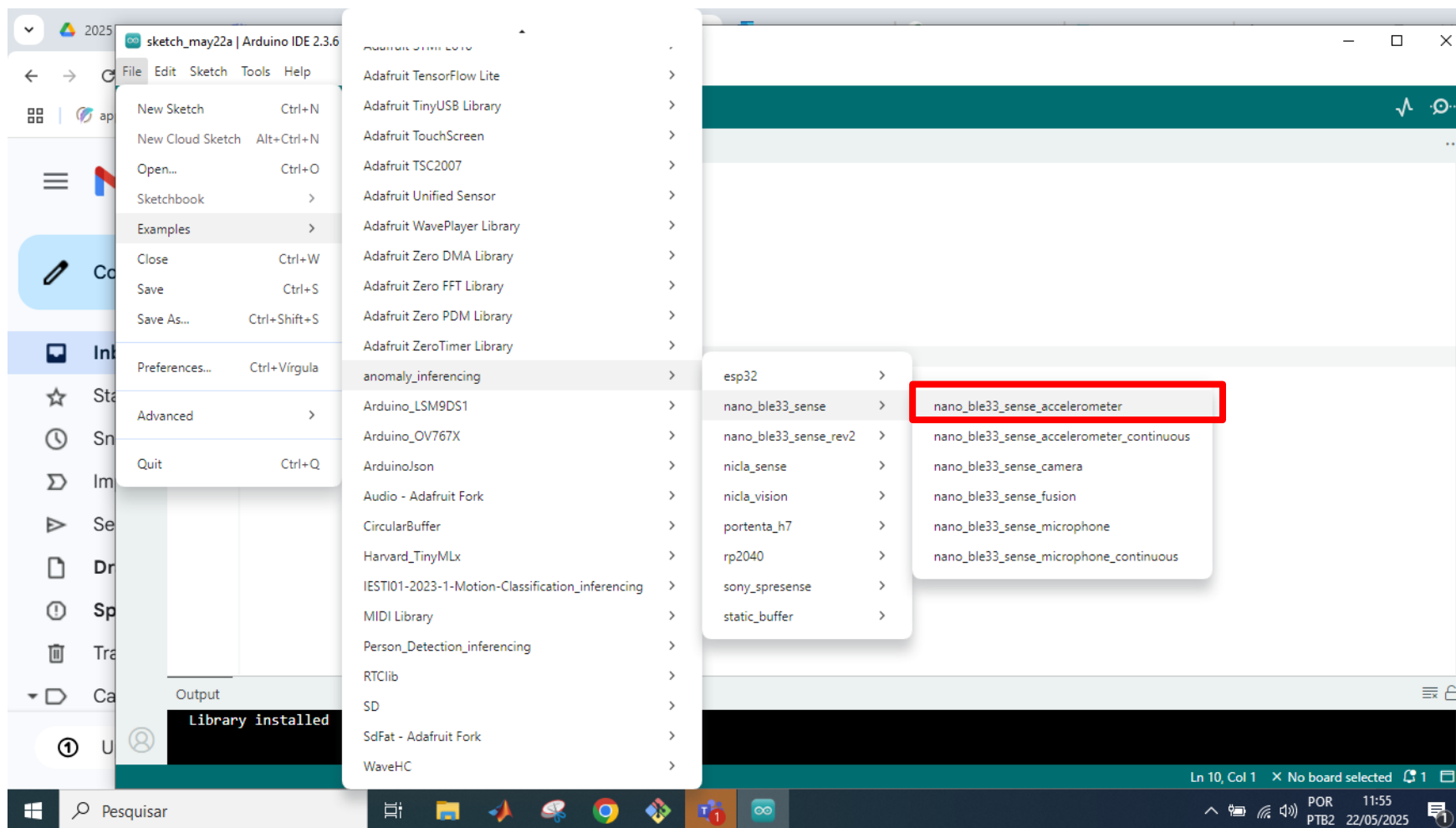
Motion Classification

8. Executando no Arduino nano-33 BLE sense



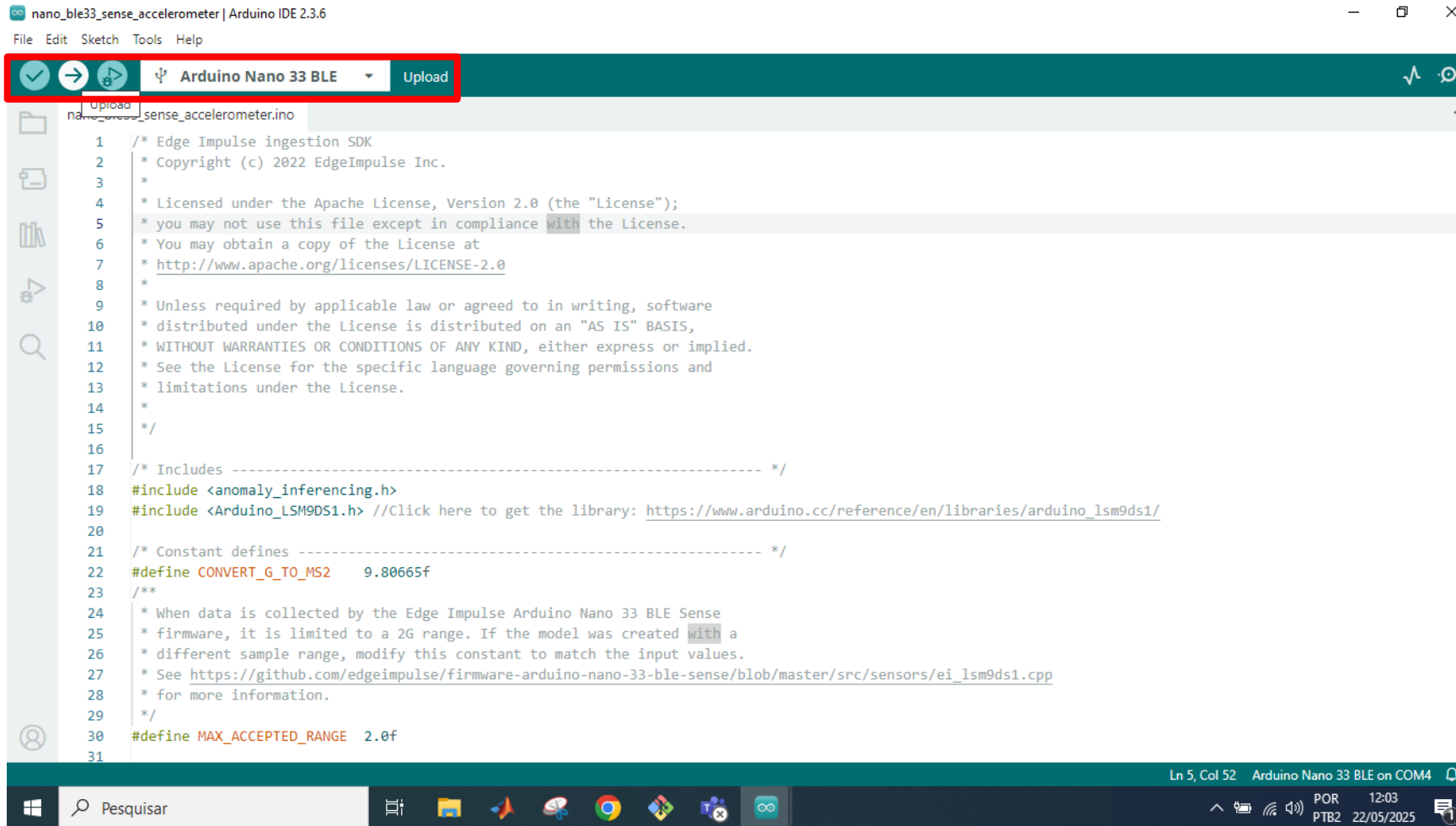
Motion Classification

8. Executando no Arduino nano-33 BLE sense



Motion Classification

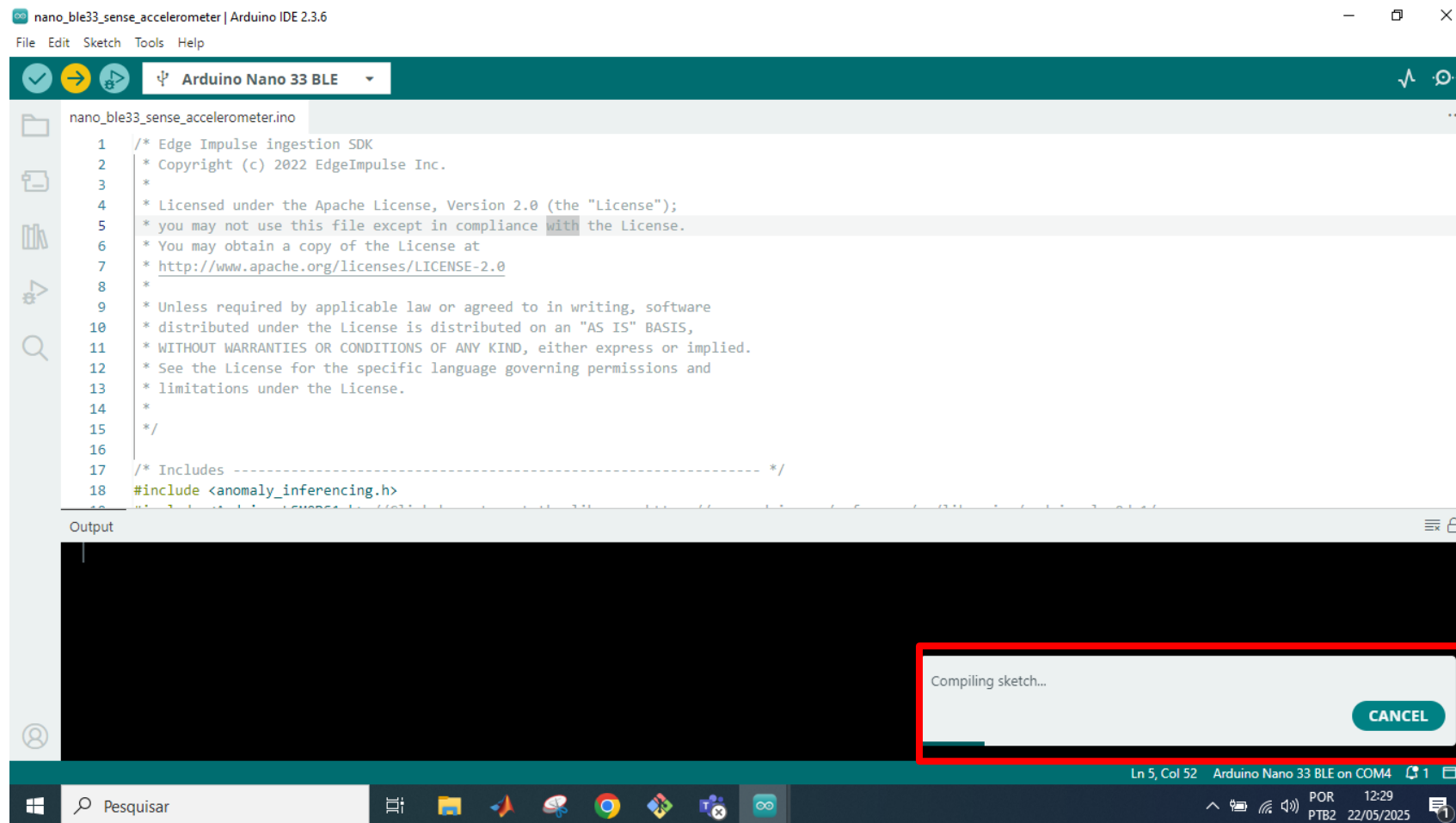
8. Executando no Arduino nano-33 BLE sense



```
1  /* Edge Impulse ingestion SDK
2  * Copyright (c) 2022 EdgeImpulse Inc.
3  *
4  * Licensed under the Apache License, Version 2.0 (the "License");
5  * you may not use this file except in compliance with the License.
6  * You may obtain a copy of the License at
7  * http://www.apache.org/licenses/LICENSE-2.0
8  *
9  * Unless required by applicable law or agreed to in writing, software
10 * distributed under the License is distributed on an "AS IS" BASIS,
11 * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
12 * See the license for the specific language governing permissions and
13 * limitations under the License.
14 *
15 */
16
17 /* Includes ----- */
18 #include <anomaly_inferencing.h>
19 #include <Arduino_LSM9DS1.h> //Click here to get the library: https://www.arduino.cc/reference/en/libraries/arduino\_lsm9ds1/
20
21 /* Constant defines ----- */
22 #define CONVERT_G_TO_MS2 9.80665f
23 /**
24  * When data is collected by the Edge Impulse Arduino Nano 33 BLE Sense
25  * firmware, it is limited to a 2G range. If the model was created with a
26  * different sample range, modify this constant to match the input values.
27  * See https://github.com/edgeimpulse/firmware-arduino-nano-33-ble-sense/blob/master/src/sensors/ei\_lsm9ds1.cpp
28  * for more information.
29  */
30 #define MAX_ACCEPTED_RANGE 2.0f
31
```

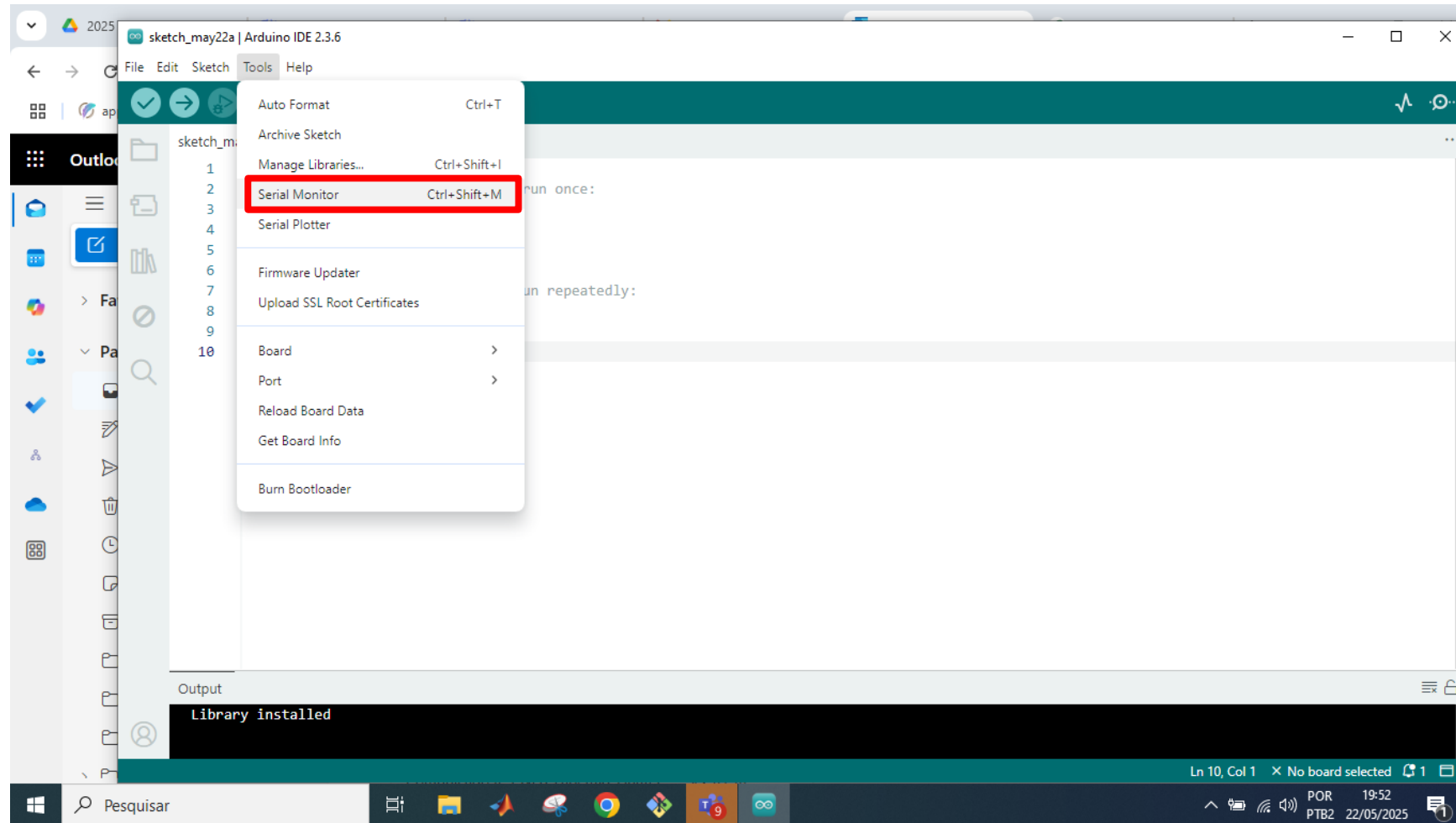
Motion Classification

8. Executando no Arduino nano-33 BLE sense



Motion Classification

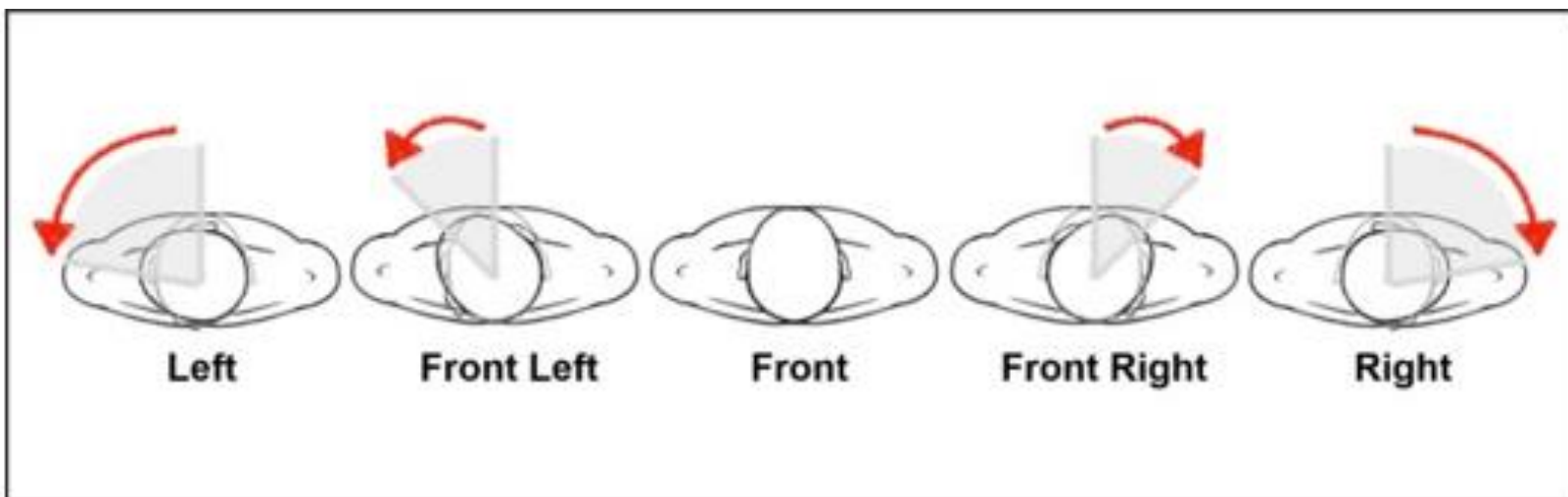
8. Executando no Arduino nano-33 BLE sense



Edge Impulse

Trabalho: Movendo uma cadeira de rodas com a cabeça

- Criar um dataset para classificação de movimento para movimentação de uma cadeira de rodas com uso do movimento da cabeça, treinar pelo menos três movimentos no Edge Impulse.
- Exemplo:



Spectral Analysis

Referências

- [1] https://github.com/Mjrovai/UNIFEI-IESTI01-TinyML/blob/main/00_Curse_Folder/2_Applications_Deploy/Class_20/EI_Studio-Spectral_Analysis-Updated_March-23.pdf
- [2] https://colab.research.google.com/github/Mjrovai/TinyML4D/blob/main/SciTinyM-2023/Edge_Impulse-Spectral_Analysis_Block/Edge_Impulse_Spectral_Analysis_Block_V3.ipynb#scrollTo=09cf0268